

VLM-driven Risk-Adaptive HUD Interactions for Trust Calibration in Automated Driving

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VLM-driven HUD parameterization for autonomous driving visualizations

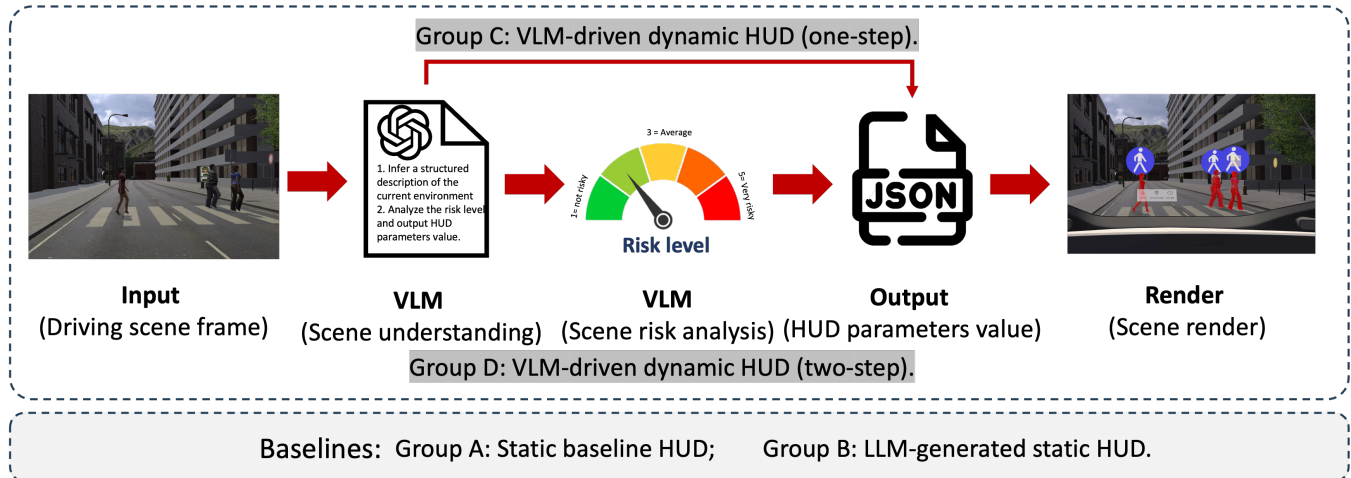


Figure 1: Overview of the proposed VLM-driven risk-to-parameter pipeline and the four HUD parameterization conditions (Groups A–D) compared in our user study.

Abstract

Automated driving trust depends on timely and intelligible feedback about driving risk. However, static “one-size-fits-all” Head-up display (HUD) designs struggle to match rapidly changing road contexts. We propose a vision-language model (VLM) approach that infers driving-scene risk and adapts HUD visualization parameters within the parameterized design space introduced by OptiCarVis. We implemented and compared four HUD parameterization strategies (static baseline; LLM-generated static; VLM-driven dynamic

HUD (one-step); and VLM-driven dynamic HUD (two-step) with explicit intermediate risk tags) in an online study (N=22). It indicated significant differences in cognitive load, trust, and perceived safety across conditions. In particular, the two-step dynamic condition yielded higher trust and perceived safety and lower cognitive load than the one-step dynamic condition, suggesting that separating risk interpretation from parameter generation can produce more interpretable adaptations.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI).**

Keywords

Automated Driving, AR-HUD, Trust Calibration, Proactive Interaction, VLM, Multimodal Explanation

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ACM Reference Format:

Mengyang Ren, Pai Zheng, and Mark Colley. 2026. VLM-driven Risk-Adaptive HUD Interactions for Trust Calibration in Automated Driving. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798997>

1 Introduction

1.1 Background

In SAE L3 automated driving, drivers remain responsible for supervising the automation and maintaining an appropriate level of engagement [2, 16]. Trust calibration and situation awareness are therefore central to safe use: users need to understand what the automated vehicle perceives and why it behaves as it does in order to monitor it effectively and avoid over- or under-reliance [17]. Interfaces should communicate driving risk in a timely and intelligible manner, supporting users' ongoing understanding of the current situation and the system's intent. However, road contexts and risks change rapidly, while many in-vehicle Head-up display (HUD) designs remain effectively "one-size-fits-all." Prior work, such as OptiCarVis [12], provides a parameterized HUD design space and evaluation dimensions, but largely targets static (or slowly updated [14]) configurations. This leaves an open question: how can we adapt HUD parameters online as risk changes, while remaining comparable within a shared design space?

In this research, a vision-language model (VLM) approach that infers driving-scene risk from ego-view frames and adapts AR-HUD visualization parameters within this design space is proposed. Our key idea is a structured risk-to-parameter mapping that can be instantiated as either one-step (direct frame-to-parameters) or two-step generation (explicit intermediate risk tags before parameter generation), enabling controlled comparisons. Figure 1 summarizes the end-to-end pipeline and the four HUD parameterization strategies evaluated in this work.

Four parameterization strategies (Groups A-D) in an online study (N=22) were evaluated using subjective measures aligned with OptiCarVis. It found that structured adaptation improves trust-related outcomes and perceived safety while managing cognitive load.

1.2 Contribution Statement:

This poster makes two contributions:

- **A structured VLM/LLM risk-to-parameter pipeline.** We introduce a risk-contingent pipeline that maps ego-view scenes to AR-HUD parameters, and operationalize both one-step and two-step generation to separate risk interpretation from parameter synthesis.
- **Empirical comparison of four parameterization strategies.** In an online study (N=22), we compare static and VLM-driven dynamic HUDs and find that the two-step dynamic approach yields higher trust and perceived safety and lower cognitive load than one-step dynamic generation, while other outcomes do not show statistically reliable differences.

2 Related work

Prior work has explored how augmented reality head-up displays (AR-HUDs) can support supervision in automated driving. OptiCarVis [12] is an important foundation that formalizes AR-HUD visualizations as a parameterized design space with evaluation dimensions aligned to user experience (e.g., trust, perceived safety, and cognitive load), enabling systematic comparison and optimization of designs. However, many solutions are static or only slowly updated [12, 14, 15], leaving open how HUD parameters should adapt *online* as road context and risk change moment-to-moment. At the same time, research on trust calibration emphasizes that interfaces should help users understand risk and anticipate system behavior [3, 5]. Risk-relevant, predictive cues can improve situation understanding and trust-related outcomes [9], while opaque or mismatched feedback can lead to inappropriate reliance [10]. Adaptive in-vehicle interfaces, therefore, often modulate both *what* is shown and *how strongly* it is emphasized based on context, uncertainty, or risk, as persistent overlays may increase distraction and workload [1, 19]. More recently, LLM/VLM-based systems connect scene understanding to interface behaviors. For example, VLM-driven methods can identify roadside risks and produce adaptive HUD cues that are perceived as useful for guiding attention [20], while LLMs can help interpret automotive UIs across design variations [7, 18]. At the same time, existing demonstrations leave key gaps for safety-critical use: dynamically generated cues are not always consistent or easily explainable, and it remains unclear how to ensure predictable parameter changes that support calibrated trust without increasing workload. Building on these directions, our work maps inferred *driving-scene risk* into a fixed, parameterized HUD space and compares one-step versus two-step generation to study how explicit intermediate risk tags affect trust, safety, and cognitive load.

3 Methodology

A VLM-driven risk-to-parameter pipeline was proposed, which infers driving-scene risk from ego-view frames and maps it to values in a parameterized AR-HUD design space (see Figure 1). The key novelty is using VLM outputs to drive HUD adaptations within a fixed parameter space (16 OptiCarVis parameters [12]), enabling controlled comparisons between two generation strategies: one-step (frame → parameters) and two-step (frame → risk tags → parameters). Concretely, the one-step VLM pipeline (Group C) directly generated HUD parameter values per keyframe (a single JSON object whose keys correspond to the 16 parameters). The two-step VLM pipeline (Group D) first generated structured environment tags and a five-level risk score (overall_risk; 1=not risky to 5=safety-critical [6]), and then generated HUD parameters conditioned on this intermediate representation; this separation is intended to produce more stable and justifiable parameter changes.

For implementation, the driving scene and HUD were built and rendered in Unity. For each 5-minute video, we saved an ego-view frame every 0.5 seconds (2Hz, 600 frames) and used the same set of frames across all conditions; only the HUD parameterization differed. We used ChatGPT-5.2 as both the VLM to interpret each

frame and the LLM to generate structured outputs. To ensure outputs remained interpretable and comparable, all generated parameter values were constrained to the ranges: Boolean parameters were treated as on/off visibility decisions, while continuous parameters (e.g., alpha and size) controlled salience, and out-of-range values were clipped to the nearest valid bound. For the dynamic strategies (Groups C and D), we updated parameters every 0.5 seconds and applied an exponential moving average smoothing method in Unity when transitioning between consecutive parameter sets to avoid abrupt jumps in element size/opacity. Parameter sequences were generated offline prior to the study and then rendered into videos in Unity.

4 User study

We conducted an online experiment via Qualtrics to compare four HUD parameterization strategies (Groups A–D) for risk communication during automated driving.

4.1 Experimental design

Conditions. Four groups (A–D) were compared, as follows:

- **Group A — Static baseline HUD.** Static HUD parameters (e.g., fixed size of the pedestrian intention cue) during automated driving. There was no change in these parameters.
- **Group B — LLM-generated static HUD.** Static HUD parameters generated by ChatGPT-5.2 and then kept fixed across scenes.
- **Group C — VLM-driven dynamic HUD (one-step).** Dynamic HUD parameters were directly generated from each ego-view frame at 0.5-second intervals using a one-step prompt that outputs only parameter values.
- **Group D — VLM-driven dynamic HUD (two-step).** Dynamic HUD parameters were generated at 0.5-second intervals using a two-step prompt: first output structured scene/risk analysis (including a five-level risk score), then output parameter values conditioned on that analysis.

Figure 2 illustrates an example of how a key HUD parameter (pedestrian intention cue size) remains constant in static designs but varies in dynamic designs.

Participants. A total of 22 participants ($M=27.57$, $SD=3.65$; 12 male and 10 female) were recruited to complete the study online. Inclusion criteria included: (i) age 18+, (ii) holding a valid driving license, and (iii) normal or corrected-to-normal vision. We collected demographic information, including gender identity, age group, education level, and employment status. Each participant watched four videos (one per condition) and completed the same questionnaire after each video. The presentation order of the four videos was randomized in Qualtrics.

Stimuli. Each stimulus was a 5-minute ego-view video of city driving in a virtual environment with a highly automated vehicle. Across videos, the driving scene and behaviors were held constant (the vehicle handled lateral and longitudinal control), and detected nearby objects were highlighted.

4.2 Measures

All measures were administered via Qualtrics after each video.

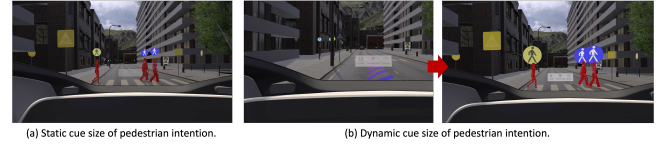


Figure 2: Example illustrating static vs. dynamic changes in pedestrian intention cue size.

- **Cognitive load.** A single-item rating of “How much mental and perceptual activity was required?” using NASA-TLX [11] on a 20-point scale (1=Very Low, 20=Very High).
- **Predictability and trust.** We adopted the Predictability/Understandability (Predictability) and Trust subscales from Körber’s Trust in Automation questionnaire [13]. Six items on a 5-point Likert scale (1=Strongly disagree, 5=Strongly agree): “The system state was always clear to me,” “I was able to understand why things happened,” “The system reacts unpredictably” (reverse-coded), “It’s difficult to identify what the system will do next” (reverse-coded), “I trust the system,” and “I can rely on the system.” We computed **predictability** as the mean of the four predictability/understandability items (including the two reverse-coded items) and **trust** as the mean of the two trust items (“I trust the system” and “I can rely on the system”).
- **Perceived safety.** A four-item semantic differential on a 7-point scale from −3 to +3: anxious–relaxed, agitated–calm, unsafe–safe, timid–confident [4, 8]. We computed **safety** as the mean of the four items.
- **Acceptance and aesthetics.** Three items on a 7-point Likert scale (1=Strongly disagree, 7=Strongly agree): “I find the visualizations of the automated vehicle useful,” “I find the visualizations of the automated vehicle satisfying,” and “I found the visualizations visually appealing.” We computed **acceptance** as the mean of the first two items (useful, satisfying) and treated **aesthetics** as the single item “visually appealing.”

5 Results

We compared conditions using Friedman tests for each outcome, followed by paired Wilcoxon signed-rank post-hoc tests with Holm correction. Kendall’s W as an effect size for omnibus tests was reported.

Friedman tests indicated significant differences across conditions for **cognitive load** ($\chi^2(3) = 10.77$, $p = .013$, $W = 0.163$), **trust** ($\chi^2(3) = 10.27$, $p = .016$, $W = 0.156$), and **perceived safety** ($\chi^2(3) = 16.44$, $p < .001$, $W = 0.249$). Differences in **predictability** ($p = .274$), **acceptance** ($p = .113$), and **aesthetics** ($p = .122$) were not significant.

Holm-adjusted Wilcoxon signed-rank post-hoc tests indicated that **Group D differed from Group C** on multiple outcomes, as shown in Figure 3: participants reported lower cognitive load ($p_{\text{Holm}} = .043$), higher trust ($p_{\text{Holm}} = .013$), and higher perceived safety ($p_{\text{Holm}} = .003$) in Group D. In addition, **perceived safety in Group D was higher than Group B** ($p_{\text{Holm}} = .006$). A pairwise difference for **acceptance** (C vs. D; $p_{\text{Holm}} = .024$) was observed, but

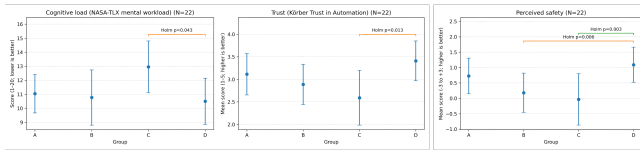


Figure 3: Summary of subjective outcomes across the four HUD conditions (A–D) in the online study (N=22).

we treat it as exploratory because the omnibus test for acceptance was not significant.

Figure 3 summarizes the mean ratings. Specifically, Group D showed the most favorable pattern overall (lower cognitive load, higher trust and safety). Group C showed the highest cognitive load and the lowest trust and safety.

6 Discussion

Overall, the results suggest that **dynamic HUD adaptation can be beneficial when it is structured**. Compared to one-step generation (Group C), the two-step approach (Group D) yielded higher **perceived safety and trust** and lower **cognitive load**. We attribute this to separating *risk-relevant scene understanding* from *visualization decisions*: the VLM first produces a structured description of what is relevant and why, and a subsequent policy maps this description to HUD parameters.

In contrast, the one-step VLM method (Group C) directly maps frames to parameters without an explicit representation of detected risks and their relevance, making parameter changes harder to interpret and more prone to inconsistent or overly reactive updates. Notably, **predictability** did not differ significantly between groups, suggesting that perceived understandability also depends on factors beyond parameter dynamics alone (e.g., cue semantics and encoding consistency). It indicated that VLM-driven interfaces should prioritize structured intermediate representations and constrained parameter policies to make dynamic adaptation usable in safety-critical contexts.

From a deployment perspective, our pipeline was implemented offline for the video-based study, but a real in-vehicle system must satisfy strict end-to-end latency constraints. Keyframes were sampled at 2 Hz and passed through a large VLM/LLM. Running such models in the loop on embedded hardware or via cloud services may introduce delays between the driving scenes and HUD parameter updates. We therefore view the current work as an upper bound on the *behavioral* benefits of dynamic adaptation under an idealized assumption of negligible generation latency.

7 Limitations and Future Work

This study used a video-based, online evaluation rather than interactive driving, which may limit ecological validity and the extent to which results generalize to real-time supervision and takeover behavior. In addition, all VLM/LLM outputs (risk tags and HUD parameter sequences) were generated offline prior to the study. We did not measure or manipulate end-to-end generation latency, nor did we investigate how delayed or jittery updates would affect user experience and trust in a real vehicle. The sample size (N=22) limits statistical power, and repeated exposure across four conditions

may introduce learning or comparison effects (partially mitigated by randomizing condition order). Methodologically, our dynamic parameter sequences were generated offline using a proprietary VLM/LLM, and sampling settings were not explicitly fixed. Future work should improve robustness across model variants, prompting choices, and diverse driving scenes. Importantly, we plan to explore *dynamic personalization*: adapting the risk-to-parameter mapping to individual drivers over time (e.g., via preference/behavioral signals) and allowing the strength and frequency of HUD adaptations to be tuned to different user needs while maintaining safety and interpretability. More importantly, future work could explore more diverse driving contexts (e.g., adverse weather conditions, or complex intersections) to improve generalizability.

8 Conclusion

We presented a VLM-driven risk-to-parameter pipeline that adapts AR-HUD visualizations within a dynamic, parameterized design space and enables controlled comparisons between one-step and two-step generation. In an online study comparing four strategies (N=22), the two-step dynamic approach (Group D) yielded higher perceived safety and trust and lower cognitive load than one-step dynamic generation (Group C), while predictability, acceptance, and aesthetics did not show statistically reliable differences. Overall, the results suggest that explicitly separating risk interpretation from parameter synthesis can support more effective risk-adaptive HUD feedback.

Acknowledgments

The authors acknowledge financial support from Smart Traffic Fund (Ref. PSRI/93/2407/PR) and Shenzhen Fundamental Research Scheme-General Program (JCYJ20230807140407016).

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